

WASP-43b

R-2708

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130212 · created 2026-09-01T00:54:53+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456335.83577 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.056
Transit depth (Rp/Rs)^2	3.74 +/- 2.16 [%]
Orbital Inclination (inc)	79.52 +/- 2.87
Airmass coefficient 1 (a1)	6.34 +/- 2.11
Airmass coefficient 2 (a2)	-1.27 +/- 1.33
Scatter in the residuals of the lightcurve fit is	36.05 %
Best Comparison Star	#1 - [580, 256]
Optimal Aperture	2.54
Optimal Annulus	6.07
Transit Duration (day)	0.038 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.056 vs 0.1594 (deviation 0.6σ)
Duration fit vs published	0.038 d vs 0.0512 d (deviation 0.74σ)
Mid-time O–C	1.1 min
Depth SNR	3.4
Residual scatter	36.05 %

Light curve

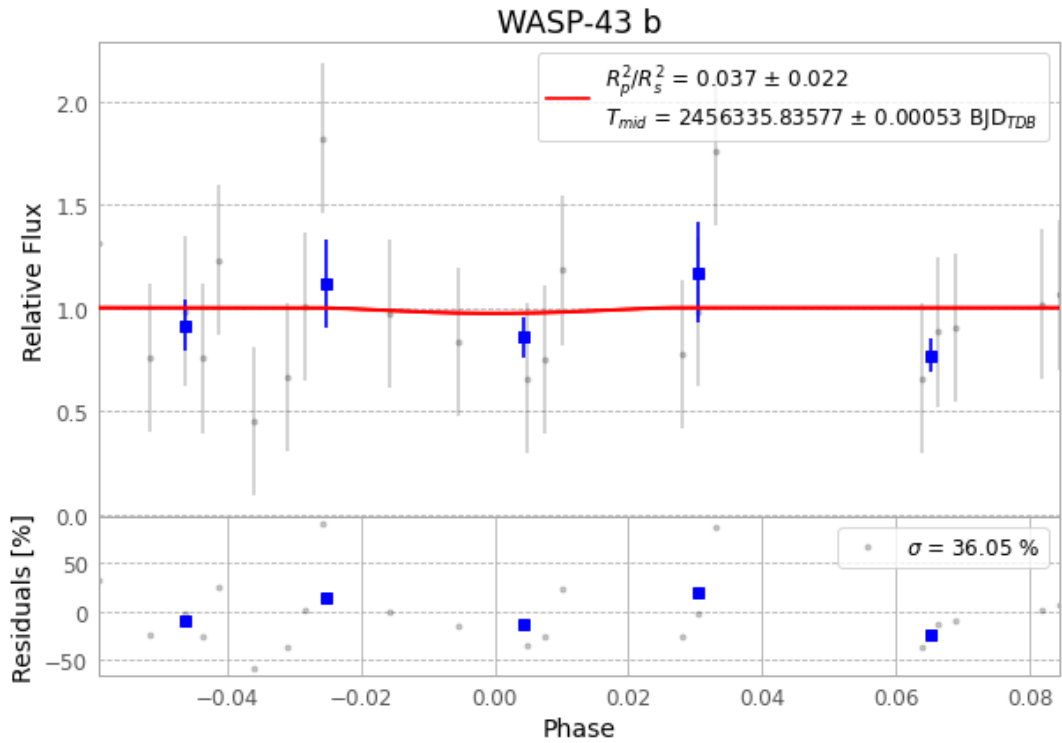


Figure 1 — FinalLightCurve_WASP-43 b_2013-02-11.png (SHA-256 f1518b47d5d99a66...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [422, 228], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7a0605bc6d049e82...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-43130212063912.FITS → WASP-43130212093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130212.log (9e08093e9ea4...), FinalLightCurve_WASP-43 b_2013-02-11.png (f1518b47d5d9...), FinalLightCurve_WASP-43 b_2013-02-11.csv (eb5e559a7175...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:54Z.